Video Demonstration-Guided Learning of Earthmoving Actions for Robotic Excavators

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***Abstract*—Excavators operate in hazardous, unstructured environments, where human operators face significant safety risks. There is clear value in introducing automation to heavy industries, such as construction, provided that the solutions are work-outcome efficient and easy to implement into the existing workflow. This paper investigates how to leverage readily accessible demonstration data to obtain useful excavator behaviors in earthmoving operations. Thus, this study proposes a three-stage framework that enables a robotic excavator to learn the manipulation of earthmoving actions from monocular video demonstrations. The framework consists of: (1) automated extraction of state-action pairs from videos, (2) initial behavioral cloning of four elementary actions (dig, swing, dump, and return), and (3) task-level optimization by training reinforcement learning policies initialized from the cloned behaviors. A simulation environment capturing soil behavior and a realistic excavator model is constructed to support training and evaluation. Experimental results demonstrate that video-based guidance can provide a relatively simple and scalable pathway toward automating heavy equipment operations.**

I. Introduction

A. Background

Excavator operations are central to commercial construction, mining, demolition, and infrastructure development. Excavators perform foundational earthmoving tasks that directly influence project timelines and overall productivity. Automating these operations can offer substantial potential benefits: it can alleviate high-skilled workforce shortages, improve productivity and task accuracy, and reduce costs. More importantly, autonomy can enhance safety by removing human operators from hazardous environments and reducing exposure to operational errors.

Despite this promise, many efforts to automate excavator operation have faced challenges in scalability and generalization. Prior work often targets a specific embodiment and a specific task environment, resulting in solutions that are difficult to transfer to new settings. Moreover, these approaches often rely on dedicated data collection, requiring excavators to be instrumented with additional sensors or operated under controlled conditions to acquire massive amounts of data for training. These requirements make the previous methodologies difficult to scale across diverse projects. As a result, deployment of autonomous excavators remains limited, and there is a growing need for methods that can leverage more easily accessible data sources while remaining adaptable to new machines and environments.

B. Problem Statement

Considering that a large quantity of lower-quality video data already exists of expert excavator demonstrations and could be cheaply collected, covering the motion of an excavator in its environment during the task of digging soil without significant locomotion. We thus consider a problem wherein we leverage this data, and at execution time, video of the ongoing earthmoving operation, without the inclusion of teleoperation information or high-quality, high-expense sensors to develop an automated excavator for a particular model that can perform this digging task. Moreover, we require that the effectiveness of the policy for this task be reasonable in terms of time, useful soil motion, and joint motion efficiency.

Therefore, this paper proposes a framework that learns control policies for excavators by using readily available monocular video data from expert demonstrations. The focus is on earthmoving operations, where cycles of four basic actions (dig, swing, dump, and return) form the core operational pattern. The proposed approach automatically extracts approximate state-action pairs from monocular video, which serve as the basis for initializing and refining autonomous behaviors. A reinforcement learning framework is then applied, initialized from the learned behaviors and further optimized for earthmoving tasks in simulation. The main contributions of this paper are:

- Automated state-action pair acquisition from monocular videos
- A video-initialized reinforcement learning framework for earthmoving operations
- A simulation environment configured for realistic excavator-soil interaction

II. Related Works

A. Automation-Augmented Excavation

Excavator automation has been an active research area for several decades, motivated by the dual imperatives of improving productivity and enhancing worker safety in hazardous environments. Azar and Kamat [17] provide a comprehensive

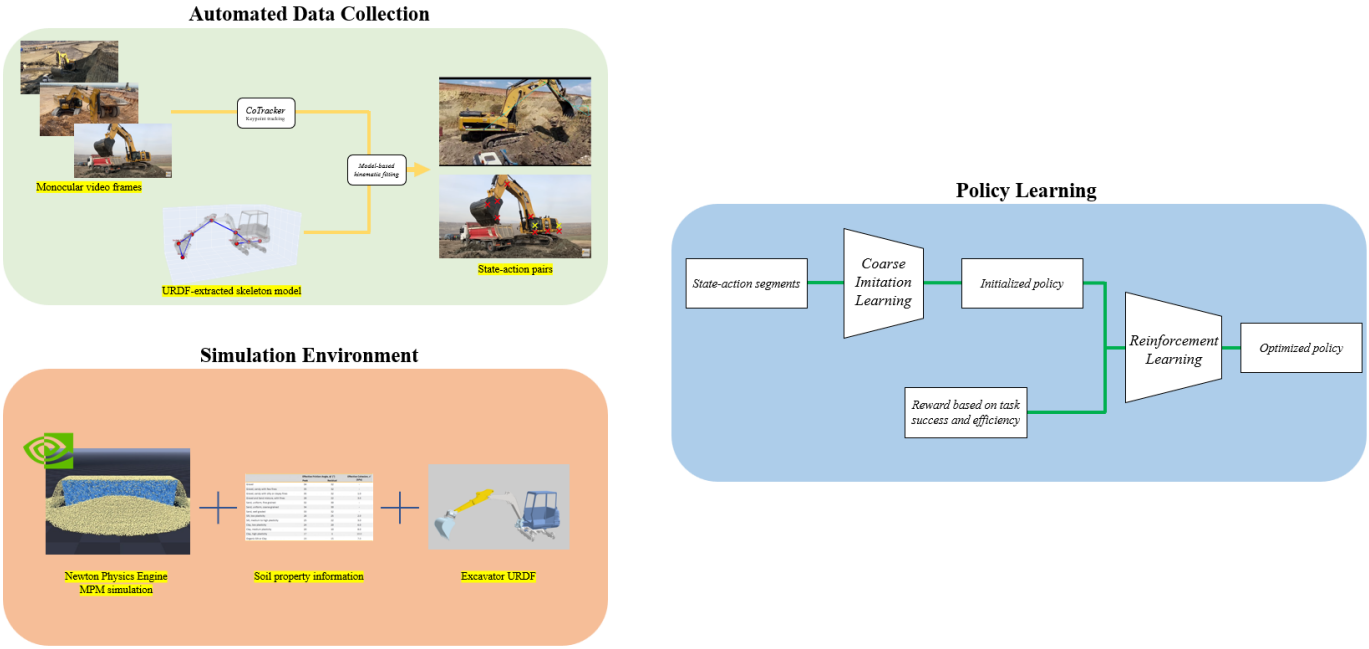


Fig. 1. Overall framework

overview of this evolution, organizing research efforts reflecting a natural progression from passive monitoring tools toward increasingly intelligent and autonomous systems.

Examples of early efforts focused on real-time localization and fleet management, using GPS [2] and local radio-frequency technologies such as RFID [16] and UWB [1] to track equipment and estimate productivity. Safety management has been another prominent area, with proximity detection and collision avoidance systems developed to protect workers from moving equipment, ranging from laser scanner-based warning systems to predictive LiDAR-based risk assessment algorithms [7]. Computer vision has become especially valuable in this context, enabling action recognition and equipment detection from site cameras [18], [14], [11].

For autonomous control of excavators themselves, remote control and teleoperation have been explored as a bridge between manual operation and full autonomy, particularly in hazardous applications. However, Azar and Kamat [17] identify this as the least developed category, with key modules still largely absent. Eraliev et al. [7] similarly conclude that the bottleneck in achieving full autonomy lies in integrating intelligent decision-making.

B. Excavation Simulators

The task of simulating excavators and their surrounding environments has already been explored in a number of directions. A variety of works seek to simulate the effects of soil interactions, primarily focusing on soil and particle motion itself and interaction schemes with rigid bodies from that perspective Zong et al. [23], Klar et al. [12]. Alternatively, they focus on reasonable ways to approximate soil for various manipulation tasks and force models for contact Holz et al. [8], Holz et al. [9], Sotiropoulos and Asada [20], Coetzee and Els [5], citetbucketfilling. Others focus primarily on modeling the motion and existence of an excavator in regards to its own dynamics such as used in the modeling in Yoo et al. [22]. Some works more closely model the relevant environment, presenting an excavator model with some manipulation capabilities in an environment with both strong interactions with deformation of the environment and in locomotion of the excavator such as in Aluckal et al. [3] and Servin et al. [19].

Our approach achieves an effective representation of reasonable soil dynamics in the close and relevant context of an automated excavator performing the digging task with a streamlined setup enabling more efficient training for reinforcement learning tasks and for the quick modification of the environment to encapsulate a more diverse context.

C. Manipulation Learning

Learning manipulation policies for excavator automation has attracted limited but growing research attention. Prior approaches have demonstrated to learn executable policies by using transformer schemes with action chunking Chen et al. [4] or by particular components using small models trained semi-separately with reinforcement learning Yoo et al. [22], Egli and Hutter [6]. However, these approaches share a dependence on rich, embodiment-specific observations, such as point cloud data or joint states directly earned from instrumented expert operation. This reliance limits the generalizability across excavator platforms and raises the barrier for data collection in practical field settings. In contrast, the proposed framework leverages broadly accessible monocular video footage of excavator operations. From these video observations, coarse action sequences are extracted and used to initialize policies

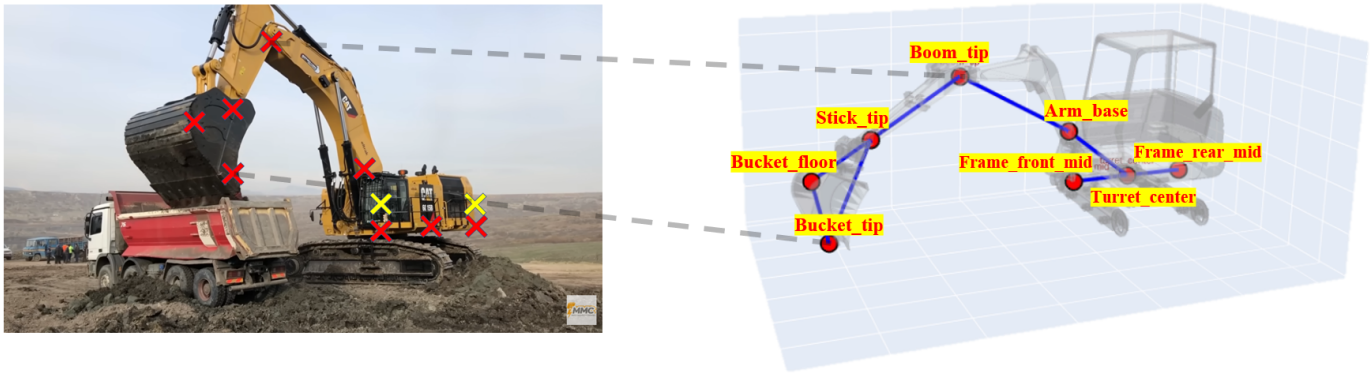


Fig. 2. (a) Tracked points in a video segment (red: eight keypoints for skeleton matching, yellow: reference points for cabin yaw estimation) and (b) skeleton extracted from URDF

via behavioral cloning. Seeded from this initialization step, a subsequent reinforcement learning refines the policies with respect to contact forces and task efficiency without requiring embodiment-specific ground truth.

III. Methods

This study proposes a three-module framework for learning earthmoving behaviors from monocular video demonstrations, as illustrated in Fig. 1. The three modules are: an automated data collection module, a simulation environment, and a policy learning module. The following sections describe each in detail.

A. Video parsing and kinematic state extraction

The pipeline transforms publicly available monocular video demonstrations of excavator operation into structured training data for behavioral cloning through a multi-stage, model-based inverse kinematics approach. Video frames are first extracted from in-the-wild footage depicting expert excavation operations. A semi-automated approach for keypoint tracking is then applied. Ten points are selected from the first frame of each video segment; eight keypoints (*frame front mid, frame rear mid, turret center, arm base, boom tip, stick tip, bucket floor, bucket tip*) used to estimate the excavator configuration in each frame, and two reference points on the excavator body, later used for cabin yaw estimation. These keypoints are defined with respect to generalized components shared across excavator families, enabling cross-embodiment applicability for both data collection and the kinematic solver. The full temporal trajectory of all ten keypoints is then recovered across the video sequence using CoTracker [10], a learned tracking model, capturing the continuous kinematic state of the excavator over time.

The kinematics solver takes ten tracked 2D image coordinates as input, and outputs the joint states of four revolute joints: three arm joints located at the arm base, boom tip, and stick tip (Fig. 2(b)), and one joint at the turret center representing cabin rotation. Based on the general motion characteristics of excavators, this study assumes all four joints move within the same vertical plane, with the plane orientation determined by the cabin yaw angle. Thus, cabin yaw estimation is performed as the first step of the kinematics solver. Four tracked points (*Frame front mid, Frame rear mid*, and the two reference points in Fig. 2) form a rectangle on the cabin surface and are used to estimate cabin yaw via homography-based planar pose tracking. The first frame of each video sequence is defined as the zero-yaw reference. The magnitude of cabin yaw is derived from the apparent width compression of the rectangle under foreshortening, and the rotation direction is determined using Eq. (1). After the cabin yaw was determined, the eight keypoints were compared to the skeleton of the URDF. The skeleton from the URDF was fitted using weak perspective projection and temporal smoothness regularization to minimize the reprojection error between the projected URDF skeleton and the annotated keypoints of the video frame. This produces a per-frame joint angle trajectory that captures the excavator's motion retargeted to the embodiment of the URDF.

$$\psi_t = \text{atan2}(\mathbf{sin}\psi_t,\ \mathbf{cos}\psi_t)\,,\ where\ |\psi| = \arccos\left(\frac{W_t}{W_0}\right) \tag{1}$$

The final step of data collection was to convert the extracted trajectories into meaningful segments. For this, four actions of excavation was recognized: dig, swing, dump, and return. The movement of the *bucket floor* and the *bucket tip* was used to determine whether the excavator was moving for dig or dump. Also, cylindrical coordinate (reach, height, and yaw) of target points for dig or dump was defined based on the local maximum point of the movement of the bucket tip. The output of this step is (θ_{cabin}, θ_{arm}, θ_{boom}, θ_{stick}, P_{dig}, P_{dump}, P_{return}), where θ represents the joint angles and P representing the cylindrical coordinates of the action target.

B. Simulation configuration

Due to the availability and general utility of the Newton simulation environment, we leverage an material point method (MPM) to approximate dirt, using Drucker-Prager results to better align groups of dirt to expectations when not in active

contact. The material of the material particles is set to values approximating reasonable dirt according to internal values. Then, a rigid body excavator is placed in the environment and simulated using a rigid body physics simulator with the components interweaved. The model for this excavator is designed to largely match a real excavator, with some extra tolerance in portions which are expected to contact particles to better account for inaccuracies in MPM simulation. The scheme is built such that target motion can be placed in the excavator every simulation step and that the model itself rejects any motion too fast or extreme for a real excavator to follow in order to better ground the behavior to reality.

C. Model architecture and design

In order to function more efficiently on limited hardware setups, the model architectures and policy representations learned from the video extracted data must ultimately be relatively lightweight. Therefore, this study explores a collection of approaches to learn the approach of the soil pile and receptacle reasonably during the dig and dump sequences. This is because the limited data (*and compute*) limits the scope of complexity applicable to the model. To overcome this limitation, the authors explored residual RL training using Soft Actor Critic (SAC) initialized with the coarse imitation learning policy, to learn the detailed corrections for contact forces and task efficiency.

Thus, we initially learn on joint information predictions of action sequences across various states in order to approximate motion in the original dataset. This design leads to building reasonable actions which effectively mimic the incoming data. However, for our scheme, we seek to generalize the task by including a reinforcement learning approach with a reward function of eq. 2, similar to that of [13], wherein we use the previous policy representation and rewards in its use as a guide to modify an added final layer which adjusts the action according to the present sequence information. Thus we continue in an RL approach to better achieve our evaluated tasks and stabilize motion. This is particularly valuable since the input data is likely to be highly varied and not quite relate to exactly the same instance of the task.

$$\boldsymbol{r}_t = \alpha \left(\frac{\boldsymbol{N}_{z>h}}{\boldsymbol{N}_{\text{total}}} \right) - \beta \boldsymbol{C}_{\text{motion}} \tag{2}$$

$\boldsymbol{\alpha}, \boldsymbol{\beta}$: Scale factors
$\boldsymbol{N}$: Number of soil particles
$\boldsymbol{C}$: Joint motion penalty by magnitude

Both for that reinforcement learning portion and in general, we are able to align our simulator tasks. Prioritized is the amount of soil successfully moved to the goal location from the start location and limited is the amount of time and motion in simulation permitted to do so, seeking more soil moved in the time frame.

IV. Experiments and Results

A. Data collection

Video frames extracted from YouTube [15] were used for the experiments at a frame rate of 10 fps. A total of 12,220 frames (*equaling to 20.4 minutes of stream*) from 10 videos were used for the experiments. These monocular videos were not collected for the purpose of robot learning, and therefore carry inherent limitations including unknown camera parameters, ambiguous excavator scale, and frequent occlusions. The proposed automated data collection framework was designed to address these challenges common to uncontrolled, in-the-wild footage by extracting a coarse information of the excavator movement itself during certain actions. Fig. 3 shows a successful extraction of dig action using the proposed framework.

An open-source excavator URDF [21] was adopted for the experiments. A skeleton of eight representative keypoints was defined as shown in Fig. 2 (b). Four of these keypoints, *stick tip*, *boom tip*, *arm base*, and *turret center* correspond directly to revolute joints in the URDF. During kinematic solving, the four points (*frame front mid*, *frame rear mid*, and the two reference points) are first configured to initialize the cabin yaw angle. In the subsequent optimization step, the remaining arm points are fitted under the assumption that all four joints lie in the same 2D plane defined by *frame front mid*, *frame rear mid*, and the two reference points (Fig. 4).

B. Model Training

Imitation learning was used to obtain a coarse initialization for the dig primitive. The authors implement six variants: a pointwise goal-conditioned MLP (1-1 MLP), two flattened-history MLPs (8-1 MLP and 16-8 MLP, where H–K denotes a history length of H states and an action chunk of K steps), a goal-conditioned encoder-decoder LSTM with scheduled teacher forcing (16-8 LSTM), an ACT with learned positional embeddings (Position Embedding ACT), and a 1-1 MLP subsequently fine-tuned with SAC in the simulation environment (1-1 MLP + RL). Each policy is trained on the same video extracted demonstrations of segments settings.

Preliminary results are summarized in Table I. The authors report two evaluation criteria: operational smoothness of the executed joint trajectory and visible soil movement progress within the simulation environment. Operational smoothness refers to the motion tending to move fluidly between distinct goal points, with a meaningful exception for cutting portions of the dig soil-contact portion and small shake outs in the dump portion. Among the pure imitation-learning variants, only the 1-1 MLP satisfies both criteria, while the higher-capacity flattened-history MLPs fail on both. We attribute this primarily to the limited volume of our current dataset, in which the additional capacity is more readily spent on the whole-body actions (*such as cabin rotation and large arm movements*) than on the underlying detailed dig motion of the bucket. The temporally-structured policies (16-8 LSTM and Position Embedding ACT) recover smoothness but do

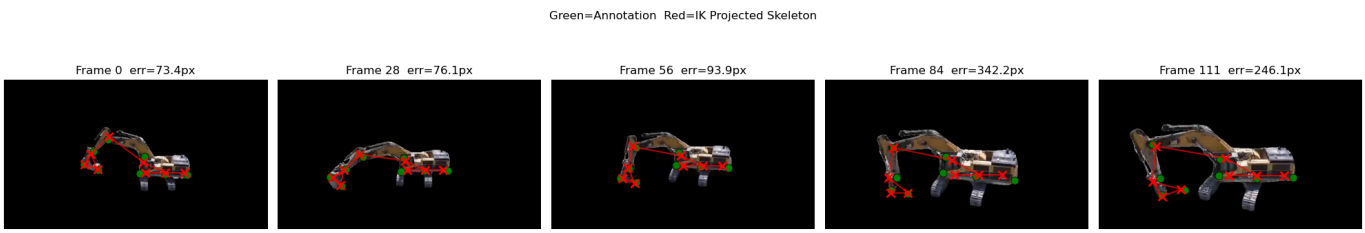


Fig. 3. Dig action result from the model-based kinematics solver



Fig. 4. A frame showing URDF skeleton overlapped after kinematic fitting. The cyan lines are the fitted skeleton.

not produce meaningful soil displacement, reflecting a gap intrinsic to video demonstrations: the recovered trajectories carry no contact-force information.

The 1-1 MLP + RL configuration satisfies both criteria (*with more distinctive soil movement than sole 1-1 MLP*), indicating that the reinforcement learning policy provides a useful correction for the coarse policy learned via MLP. However, as shown in Fig. 5, the excavator throws the soil into the air to increase the return on rewards. We denote this as a limitation of the simulation tool, where the granular particles by the MPM solver leaks out of the corner of the bucket's mesh on finer settings. The authors expect that solving this issue of simulation compatibility will improve the RL fine-tuning to learn more meaningful dig actions.

Model	Smoothness	Soil Movement Progress
1-1 MLP	X	X
8-1 MLP		
16-8 MLP		
16-8 LSTM	X	
Position Embedding ACT	X	
1-1 MLP + RL	X	X

TABLE I
COMPARISON OF MODEL PERFORMANCE.

Additionally, Fig. 6 shows the dump action produced by the imitation learning policy. RL finetuning of dumping was not attempted due to a current limitation of the soil simulator. Even if the excavator scooped some soil during the dig phase, the particles leaked from the bucket throughout the subsequent swing phase. Thus, when the excavator proceeded to a dump step, no soil was left to evaluate the reward. However, the imitation policy produces a kinematically plausible dumping motion, indicating that the primitive is learnable once the simulator issue is resolved.

C. Simulation

Qualitatively, after considerable stress testing, we find that the excavator model behaves extremely reasonably. Although it is possible for the excavator to reach effectively irrecoverable states, such as falling over, doing so accidentally or having some further reaction is, if possible, extremely difficult to achieve.

Similarly, in a number of different configurations (Fig. 7), we find that the soil simulation itself appears stable and the particles move reasonably.

We acknowledge however two limitations. First, the interactions between particles and the rigid body can on occasion be unreasonable; the particles leak out when interacted with the rigid body of the excavator. This limitation critically affects learning in the RL process as it cannot hold onto soil for a long time. Second, the simulation itself can be somewhat slow. Indeed, the simulation on our present hardware involves sufficient calculation that, even approximated, it runs at slower-than-real-time rates. This may make the RL approach later difficult, but it may either be the case that the delay is not overwhelming for our training purposes or that portions of the simulation can be optimized out as we move forward.

V. Discussions and Conclusion

The experimental results demonstrate that a video-derived imitation baseline, when paired with reinforcement learning in a physics-accurate soil simulation, has the potential to produce an excavator policy that executes smoothly and makes measurable progress on the dig task. The comparative study further shows that, in the current data regime (*a limited amount of data*), architectural capacity is not the dominant factor; the pointwise 1-1 MLP outperforms its higher-capacity counterparts under pure imitation, and the temporally-structured LSTM and ACT variants recover trajectory smoothness but not soil engagement. This suggests that the primary obstacle is the scaling of demonstration data.

The limitations of the current study follow directly from this observation. The demonstration set remains small (*12,200 frames representing 20 min of video*) and carries residual noise from the homography-based kinematics fitting. It is also

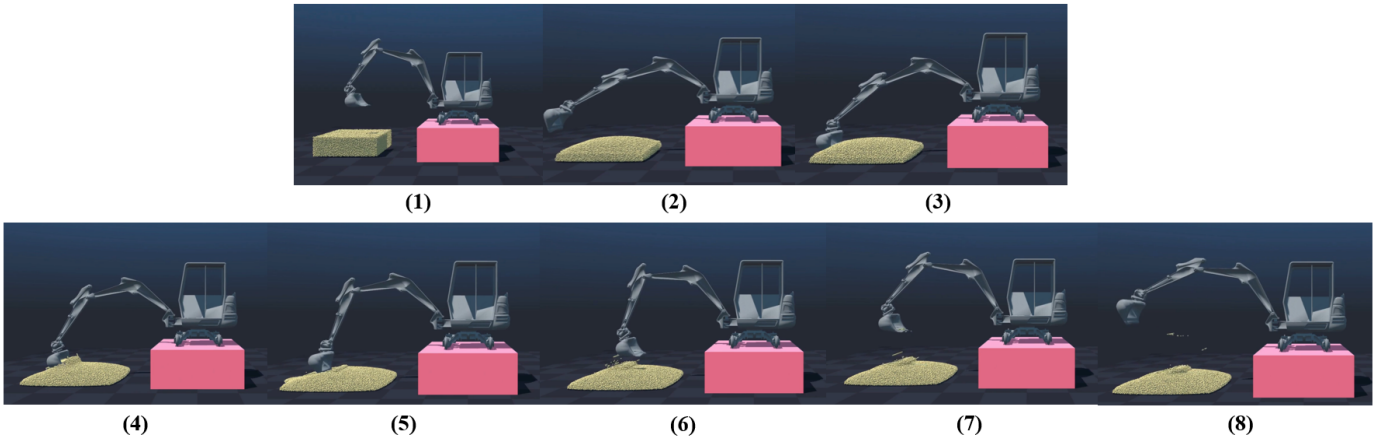


Fig. 5. Result of a learned dig policy in order of time. (1)-(3) shows dig target approach via BC initialization and (4)-(8) shows the excavator trying to dig via RL policy

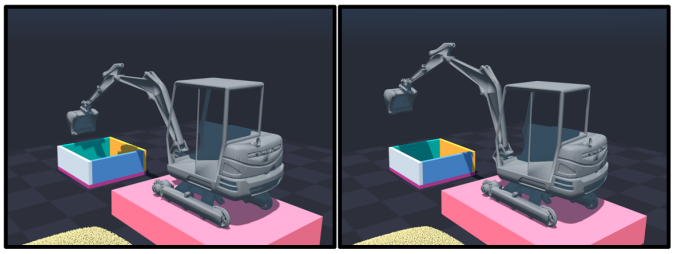


Fig. 6. Dump action (left to right) from BC policy. The soil did not follow due to simulation issues.

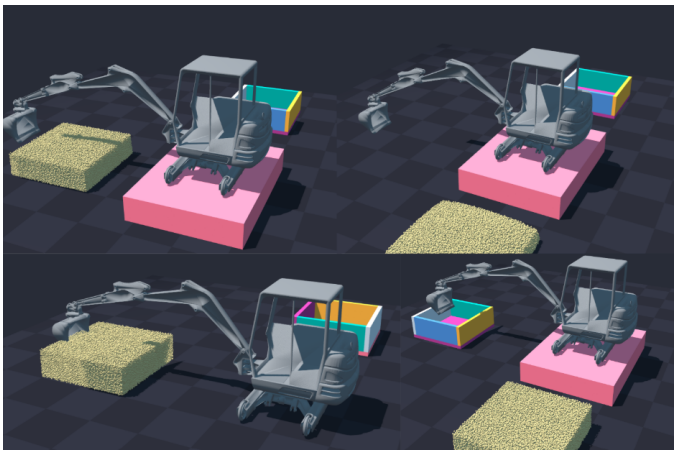


Fig. 7. Different configurations of the simulation environment

challenging to extract any kind of context information from the video itself, as the dynamics of the granular soil is arduous to understand from video alone. The imitation baselines are therefore best interpreted as coarse initializations rather than standalone controllers. Another limitation is the reinforcement-learning side, where the pipeline is currently restricted to the dig primitive, and the reward shaping treats soil displacement and smoothness as separable objectives rather than coupled costs.

A. *Future work*

Thus, future work falls naturally into three directions, each tied to the limitations above. First, we plan to scale the demonstration dataset by automating the keypoint extraction pipeline and reducing reliance on manual verification, which should improve both the absolute quality of the recovered joint trajectories and the statistical coverage of the action distribution. We expect this to disproportionately benefit the higher-capacity and temporally-structured imitation variants, which are currently starving of data. Second, we plan to extend the RL fine-tuning stage beyond the dig primitive to the full dig–swing–dump–return sequence, including phase transitions, and to generalize across a range of soil parameters and bucket entry configurations. Third, we intend to tighten the coupling between the imitation and reinforcement learning stages by treating the coarse policy as a behavioral prior during RL (e.g., through KL regularization) rather than as a one-shot initialization, which should allow the learned policy to retain demonstration-like kinematics while adapting to contact dynamics.

B. *Conclusion*

We presented a pipeline for learning fundamental excavator actions from video demonstrations, combining a homography-based kinematic solver, a family of goal-conditioned imitation policies, and reinforcement-learning fine-tuning in a simulator with MPM solvers for granular soil movement. A comparative study across six policy variants shows that, under limited and noisy video-derived supervision, a simple pointwise imitation policy provides the most reliable initialization, and that the remaining kinematic-to-dynamic gap can be closed by reinforcement learning against the full soil physics.

The contributions of this work are threefold. First, in contrast to prior excavator learning studies that rely on proprietary data sources tightly coupled to a specific embodiment, such as on-board LiDAR streams or joint-level telemetry collected through teleoperation, our pipeline draws supervision exclusively from publicly available video footage, dramatically

lowering the barrier to demonstration collection. Second, we propose a data collection and retargeting procedure that extracts kinematic trajectories from video and projects them onto a model-specific skeleton, which in principle enables cross-embodiment data collection across different excavator families from the same underlying video corpus. Third, to our knowledge, the use of a residual reinforcement learning pipeline is novel within the excavator learning literature, and it directly addresses a fundamental limitation of video-only supervision by recovering the contact dynamics that purely kinematic demonstrations cannot convey.

More broadly, we view the proposed framework as applicable beyond the excavator setting: any manipulator-like machines whose actuated joints move predominantly within a single plane embedded in three-dimensional space admits the same homography-based recovery and retargeting strategy, suggesting that the pipeline may transfer to a wider class of planar-articulated machines. While the current results are not yet of a grade suitable for real-world deployment, they establish the feasibility of the overall approach and identify dataset scale, primitive coverage, and imitation–RL coupling as the most productive targets for subsequent work.

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